IT4010 Research Project — 2026 May

End of Month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient loT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	
Research Cluster	Software Systems & Technologies (SST)
Specialization	
Submission Period	16th May – 30th May 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	55%

Section C — Tasks This Period

C1. Tasks Completed

- Finalized and solidified the literature review following PP1 examiner feedback, consolidating 12 peer-reviewed references in IEEE format and repositioning the contribution against Giordano et al. (ENSsys'23) as a joint multi-output extension.
- Incorporated PP1 viva feedback into the research documentation and confirmed the four-output adaptive policy (T*, ΔF*, U, AoI) as the core research contribution.
- Continued refinement of the ESP32 adaptive FSM firmware: re-characterized MQ-135 and MQ-5 gas sensor warmup behaviour and tuned alert thresholds above the confirmed idle baseline to eliminate false ALERT triggers.
- Conducted controlled comparison runs between BASELINE (fixed 60-second) and ADAPTIVE transmission modes, capturing preliminary Transmission Count Reduction (TCR) readings through the live dashboard.
- Began Raspberry Pi edge gateway setup (WBS-4): initial environment configuration and Mosquitto broker installation in preparation for migrating the gateway feedback loop from the laptop proxy.
- Updated and documented the project repository with the latest firmware, backend, and dashboard code; maintained research logbook entries covering the post-PP1

implementation work.

- Held an informal check-in with the supervisor to confirm post-PP1 direction and align on Progress Presentation 2 planning and deliverables.

C2. Tasks Planned for Next Period

- Complete the Raspberry Pi edge gateway: migrate the FSM gateway feedback loop from the laptop proxy and implement cluster-level priority arbitration (WBS-4).
- Implement composite urgency scoring (U = w1F + w2G) and OVERRIDE_T feedback from the gateway to individual bin nodes.
- Begin mobile application development using React Native with Expo: collection schedules, push alerts, and issue reporting (WBS-7).
- Conduct full quantitative evaluation: measure TCR, Alert Recall (AR), and Battery Life Extension Factor (BLEF) against the fixed-interval baseline using a USB power meter (WBS-9).
- Complete Status Document 1 and continue bi-weekly research logbook entries.
- Schedule the next formal supervisor meeting to review edge gateway progress and begin Progress Presentation 2 preparation.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- Migrating the laptop-based gateway logic to the Raspberry Pi requires re-validating the MQTT feedback loop timing; plan to test incrementally and confirm round-trip latency before integration.
- Capturing reliable energy measurements requires consistent test conditions; plan to standardize the fill profile and use a USB power meter for repeatable TCR and BLEF readings. Workload across final-year modules continues to be managed with dedicated IT4010 time blocks.

Section E — Supervisor Meeting Notes

Date of Meeting	Late May 2026 (informal check-in)
Mode	Informal check-in

Key Points Discussed

- Reviewed PP1 outcome and examiner feedback; confirmed the finalized literature review and the four-output adaptive policy as the core contribution.
- Confirmed post-PP1 direction: proceed with Raspberry Pi edge gateway implementation and prepare for quantitative evaluation.
- Agreed that PP2 deliverables will include the edge gateway, mobile app, cluster

simulation, and measured TCR / AR / BLEF results.

- Next formal supervisor meeting to be scheduled in June to review edge gateway progress.

Section F — Research Logbook Reference

Logbook Entry Dates	
Page / Entry Numbers	

Student Declaration

I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____